

Maxima Minima

Derivatives

$$f(x) = x^n$$

$$f'(x) = \frac{d}{dx} f(x)$$

$$= \frac{d}{dx} x^n$$

$$= n x^{n-1} \frac{dx}{dx}$$

$$= n x^{n-1}$$

Partial Derivative

$$f(x) = x^n$$

$$f'(x) = \frac{d}{dx} x^n$$

$$= n x^{n-1} \frac{d(x)}{dx}$$

$$= n x^{n-1}$$

Example of Derivatives

$$f(x) = x + y + xy$$

$$f'(x) = \frac{d}{dx} f(x)$$

$$= 1 + y' + xy' + y(1)$$

Example of Partial Derivative

$$f(x) = x^n$$

$$f'(x) = \frac{\partial}{\partial x} x^n$$

$$= nx^{n-1} \frac{\partial}{\partial x} (x)$$

$$f(x) = x + y + xy$$

$$= \frac{\partial}{\partial x} (x + y + xy)$$

$$= 1 + 0 + y$$

Example $f(x, y) = x^3 + y^3 + 3xy$

$$f_x = 3x^2 + 0 + 3y$$

$$f_y = 0 + 3y^2 + 3x$$

$$f_{xx} = 6x + 0 + 0$$

$$f_{yx} = 0 + 0 + 3$$

$$f_{xy} = 0 + 0 + 3$$

$$\therefore a^3 + b^3 = (a+b)(a^2 + ab + b^2)$$

$$\textcircled{1} \quad f_{xx} f_{yy} - (f_{xy})^2 \quad \begin{array}{l} y > 0 \quad \text{Min} \\ y < 0 \quad \text{Max} \end{array}$$

$$(6x)(6y) - 9$$

$$\textcircled{1} \quad f_x = 0 = 3x^2 + 3y = x^2 + y$$

$$\textcircled{2} \quad f_y = 0 = 3y^2 + 3x = y^2 + x$$

$$\text{from } \textcircled{1} \quad y = -x^2$$

put it in eq (2)

$$x + (-x^2)^2 = 0$$

$$x + x^4 = 0$$

$$x(1 + x^3) = 0$$

$$x = 0 \quad (1+x)(1+x+x^2) = 0$$

$$x = 0 \quad x = -1 \quad 1+x+x^2 \in \mathbb{C} \text{ complex}$$

$$\text{if } x = 0 \quad y = 0$$

$$(0, 0)$$

$$y = 0 \quad y = 0$$

$$\text{if } x = -1 \quad (-1, -1)$$

$$y = -1$$

$$6x \cdot 6y - 9$$

$$6(0) \cdot 6(0) - 9$$

$$0 - 9$$

$$-9 < 0 \quad \text{Max}$$

$$-9 < 0 \quad \text{Max}$$

$$Q) f(x, y) = x^4 - y^4 + 3xy \quad \therefore f_{xx}f_{yy} - (f_{xy})^2$$

$$f_x = 4x^3 - 0 + 3y$$

$$f_y = 0 - 4y^3 + 3x$$

$$f_{xx} = 12x^2 - 0 + 0$$

$$f_{yy} = f_{yy} = -12y^2$$

$$f_{xy} = 0 - 0 + 3$$

$$f_{yx} = 0 - 0 + 3$$

$$f_{xx}f_{yy} - (f_{xy})^2$$

$$(12x^2)(-12y^2) - (3)^2$$

$$f_x = 0 = 4x^3 + 3y \quad \text{--- (1)}$$

$$f_y = 0 = 3x - 4y^3 \quad \text{--- (2)}$$

from (1) ~~3x~~

$$y = -\frac{4}{3}x^3$$

put in eq 2

$$-4 \left(-\frac{4}{3}x^3 \right)^3 + 3x = 0$$

$$\frac{256 x^9 + 3x}{27} = 0$$

$$256 x^9 + 81 x = 0$$

$$x = 0, 256 x^8 + 81 = 0$$

$$\text{let } x^4 = a$$

$$(16 x^4)^2 + 2(16 x^4)(9) + (9)^2 = 2(16)(x^4)(9)$$

$$\text{let } x^2 = a$$

$$(16 x^4 + 9)^2 = 288 x^4$$

$$(16 x^4 + 9)^2 - (\sqrt{288} x^2)^2 = 0$$

$$\text{BiQuadra } (16 x^4 + 9 - \sqrt{288} x^2)(16 x^4 + 9 + \sqrt{288} x^2) = 0$$

$$16 x^4 - \sqrt{288} x^2 + 9 = 0 ; 16 x^4 + \sqrt{288} x^2 + 9 = 0$$

$$16 a^2 - \sqrt{288} a + 9 = 0 ; 16 a^2 + \sqrt{288} a + 9 = 0$$

$$x_1 = \frac{3\sqrt{2}}{8} + \frac{3\sqrt{2}i}{8}$$

$$x_2 = \frac{3\sqrt{2}}{8} - \frac{3\sqrt{2}i}{8}$$

$$x_1 = \frac{-3\sqrt{2}}{8} + \frac{3\sqrt{2}i}{8}$$

$$x_2 = \frac{3\sqrt{2}}{8} - \frac{3\sqrt{2}i}{8}$$

$$a = \frac{3\sqrt{2}}{8} \pm \frac{3\sqrt{2}i}{8} \in \mathbb{C} \quad a = \frac{3\sqrt{2}}{8} \pm \frac{3\sqrt{2}i}{8} \in \mathbb{C}$$

$$x^2 = a$$

$$x^2 = (a \pm bi)^2$$

$$a^2 \pm 2abi - b^2 \in \mathbb{C}$$

if $x=0$

$$y = -\frac{9}{3} x^3$$

$$y = 0$$

put $(0, 0)$

$$(12x^2)(-12y^2) - (3)^2$$

$$(12(0)^2)(-12(0)^2) - 9$$

$$-9 < 0 \quad \text{Max}$$

Multi function / Multi variable chain rule

Suppose $y^t = \left(4 + \underbrace{(x^2 + 1)}_u\right)^3$

$$\frac{\partial y}{\partial t} = t^3$$

$$\frac{\partial u}{\partial x} = x^2 + 1$$

$$\frac{\partial t}{\partial u} = 4 + 4$$

$$\frac{\partial y}{\partial x} = \frac{\partial y}{\partial t} \times \frac{\partial t}{\partial u} \times \frac{\partial u}{\partial x}$$

$$\frac{\partial y}{\partial x} = \frac{\partial y}{\partial u} \cdot \frac{\partial u}{\partial x}$$

Q) $f(x, y, z) = x^2y + \cos y + y \sin x$

$f_x = 2xy + 0 + y \cos x$

$f_{xx} = 2y + 0 + y \sin x$

$f_y = x^2 - \sin y + \sin x$

$f_z = 0$

Q) 14.2

$\lim_{(x,y) \rightarrow (0,0)} \frac{3x^2 - y^2 + 5}{x^3 + y^2 + 2}$

Applying limit

$= \frac{3(0)^2 - (0)^2 + 5}{(0)^3 + (0)^2 + 2}$

$= \frac{5}{2}$ Ans

Q) $\lim_{(x,y) \rightarrow (0,0)} \frac{e^y \sin x}{x}$

$\lim_{(x,y) \rightarrow (0,0)} \frac{e^y \sin x}{x}$

$\therefore \frac{\sin x}{x} \approx 1$

Applying limit

$e^0 \times 1 = 1$

Q) $\lim_{(x,y) \rightarrow (0,0)}$

$$Q) \lim_{(x,y) \rightarrow (0,0)} \frac{1 - \cos xy}{xy}$$

$$= \frac{1 - \cos xy}{xy} \times \frac{1 + \cos xy}{1 + \cos xy}$$

$$= \frac{1 - \cos^2 xy}{xy (1 + \cos xy)}$$

$$= \frac{\sin^2 xy}{(xy) (1 + \cos xy)}$$

$$= \left(\frac{\sin xy}{xy} \right) \left(\frac{\sin xy}{1 + \cos xy} \right)$$

Applying limit

$$= \frac{1 \times 0}{2}$$

$$= 0$$

$$Q) \lim_{(x,y) \rightarrow (1,1)} \frac{x - y + 2\sqrt{x} - 2\sqrt{y}}{\sqrt{x} - \sqrt{y}}$$

$$\lim_{(x,y) \rightarrow (1,1)} \frac{\{(\sqrt{x})^2 - (\sqrt{y})^2\} + 2(\sqrt{x} - \sqrt{y})}{\sqrt{x} - \sqrt{y}}$$

14.3

$$\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} ?$$

$$f(x,y) = 2x^2 - 3y - 4$$

$$f_x = 4x$$

$$f_{xx} = 4$$

$$f_y = -3$$

$$f_{yy} = 0$$

14.6 (27-30)

Linearization

$$L(x,y) = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$

Q.27

$$f(x,y) = x^2 + y^2 + 1 \quad \text{at } (0,0) \text{ and } (1,1)$$

$$f_x = 2x \quad \text{at } (0,0) \Rightarrow f_x = 0$$

$$f_y = 2y \quad \text{at } (0,0) \Rightarrow f_y = 0$$

Now

$$= f(0,0) + f_x(0,0)(x-0) + f_y(0,0)(y-0)$$

$$= 1 + 0(x-0) + 0(y-0)$$

$$= 1$$

Now at $(1,1)$

$$f_x = 2x \quad \text{at } (1,1) = 2$$

$$f_y = 2y \quad \text{at } (1,1) = 2$$

$$L(x,y) = f(1,1) + f_x(1,1)(x-1) + f_y(1,1)(y-1)$$

$$= 1^2 + 1^2 + 1 + 2(x-1) + 2(y-1)$$

$$= 3 + 2x - 2 + 2y - 2$$

$$= 2x + 2y - 1 \quad \underline{\text{Ans}}$$

14-6 (1-6)

Tangent plane & Normal line

Tangent

$$f_x(P_0)(x-x_0) + f_y(P_0)(y-y_0) + f_z(P_0)(z-z_0) = 0$$

Normal line

$$x = x_0 + f_x(P_0)t$$

$$y = y_0 + f_y(P_0)t$$

$$z = z_0 + f_z(P_0)t$$

$$\text{Q.1) } x^2 + y^2 + z^2 = 3 \quad P_0(x_0, y_0, z_0) = (1, 1, 1)$$

$$f_{x_{P_0}} = 2x \text{ at } (1, 1, 1) = 2$$

$$f_y = 2y \text{ at } (1, 1, 1) = 2$$

$$f_z = 2z \text{ at } (1, 1, 1) = 2$$

$$f_x(P_0)(x-x_0) + f_y(P_0)(y-y_0) + f_z(P_0)(z-z_0) = 0$$

$$2(x-1) + 2(y-1) + 2(z-1) = 0$$

$$2x + 2y + 2z - 6 = 0$$

Normal line

$$x = 1 + 2t, \quad y = 1 + 2t, \quad z = 1 + 2t$$

$$f_x(P_0)t = x - x_0$$

$$f_y(P_0)t = y - y_0$$

$$f_z(P_0)t = z - z_0$$

if $a \rightarrow 0$ then Taylor Series

Taylor Series 14.9 (1-5)

One variable

$$\frac{1}{0!} f(a) (x-a)^0 + \frac{1}{1!} f'(a) (x-a)^1 + \frac{1}{2!} f''(a) (x-a)^2$$

$$+ \frac{1}{3!} f'''(a) (x-a)^3 \dots$$

More than one variable at (origin)

$$\frac{1}{0!} f(0,0) + \frac{1}{1!} \{ x f_x + y f_y \} \quad \left. \vphantom{\frac{1}{1!} \{ x f_x + y f_y \}} \right\} \text{Quadratic}$$

$$+ \frac{1}{2!} \{ x^2 f_{xx} + 2xy f_{xy} + y^2 f_{yy} \}$$

$$+ \frac{1}{3!} \{ x^3 f_{xxx} + 3x^2y f_{xxy} + 3xy^2 f_{xyy} + y^3 f_{yyy} \} \quad \text{cubic}$$

$$+ \frac{1}{4!}$$

$$+ \frac{1}{5!}$$

$$\iiint x^3 dx = \frac{x^4}{4}$$

$$\int x^3 dx dy dz = \frac{x^4}{4} yz$$

$$\int x dy = xy + c$$

$$(x+y)' = x f_x + y f_y$$

$$(x+y)^2 = x^2 f_{xx} + 2xy f_{xy} + y^2 f_{yy}$$

$$(x+y)^3 = x^3 f_{xxx} + 3x^2y f_{xxy} + 3xy^2 f_{xyy} + y^3 f_{yyy}$$

Q.1) $f(x, y) = x e^y$

$$f_x = e^y; f_{xx} = 0; f_{xxx} = 0$$

$$f_y = x e^y; f_{yy} = x e^y; f_{yyy} = x e^y$$

$$f_{xy} = e^y, f_{xxy} = 0$$

$$f_{xyy} = e^y$$

$$\frac{1}{0!} (0 \times e^0) + \frac{1}{1!} \{ x e^y + y x e^y \} + \frac{1}{2!} \{ x^2 (0) \}$$

$$+ \frac{1}{2!} \{ x^2 (0) + 2xy e^y + y^2 x e^y \}$$

~~+~~

$$+ \frac{1}{3!} \{ x^3 (0) + 3x^2 y (0) + 3xy^2 (e^y) + y^3 (x e^y) \}$$